

Data Flow Diagram

Date	26 june 2026
Team ID	xxxxxx
Project Name	Opticrop-Smart Agricultural Production Optimization Engine”
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	Farmer / User: Sends soil data and environmental factors; receives the final crop recommendation.
Rectangle with numbered header	Process	1 Input Validation & Handling: What it does: Takes the raw values entered by the farmer (N, P, K, pH, temperature, humidity, rainfall) and ensures they are valid numbers. 2 Crop Prediction Engine: What it does: Passes the validated inputs into your trained Machine Learning model (like Random Forest) to calculate the best crop match. 3 Results Generation & Display: What it does: Formats the prediction into a user-friendly recommendation webpage.
Rectangle (solid fill, no number)	Data Store	Crop Model Registry (crop_model.pkl): Holds the pre-trained machine learning model weights used by the prediction engine.

		User History Logs (Optional Data Store): Stores past search queries and recommended crops for future reference.
Labeled arrow	Data Flow	Crop Model Registry (crop_model.pkl): Holds the pre-trained machine learning model weights used by the prediction engine. User History Logs (Optional Data Store): Stores past search queries and recommended crops for future reference.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

